

EXPERIMENT 1

Bluetooth

Bluetooth is a short-range wireless technology operating in the 2.4 GHz ISM band, using Frequency-Hopping Spread Spectrum (FHSS) for interference avoidance. It supports data rates up to 5 Mbps (Bluetooth 5.0) and has a range of 10 to 100 meters, depending on the class. Bluetooth Low Energy (BLE) is ideal for low-power applications like wearables and IoT devices.

Wi-Fi

Wi-Fi is used for wireless networking in local area networks (LANs), typically operating on 2.4 GHz, 5 GHz, and 6 GHz bands. Wi-Fi 6 offers data rates up to 9.6 Gbps and provides coverage of 30 meters indoors. It's widely used for high-speed internet, streaming, and connecting multiple devices.

Zigbee

Zigbee is a low-power, low-data-rate wireless protocol designed for IoT and automation. It operates in the 2.4 GHz, 868 MHz, and 915 MHz bands and supports mesh networking. Its data rate is up to 250 kbps, with a range of 10 to 100 meters. It's commonly used in home automation and industrial sensor networks.

LoRa

LoRa is a long-range, low-power wireless technology for IoT applications. Operating in sub-GHz ISM bands (433 MHz, 868 MHz, 915 MHz), it uses Chirp Spread Spectrum (CSS) for robust communication over long distances. Data rates range from 0.3 kbps to 27 kbps, with ranges up to 15 km in rural areas.

5G

5G is the fifth generation of mobile communication, offering ultra-fast speeds and low latency. It uses a mix of sub-6 GHz and millimeter-wave bands. Theoretical speeds reach 20 Gbps, with practical speeds up to 10 Gbps. It enables applications like autonomous vehicles, smart cities, and industrial automation.

NFC

NFC operates at 13.56 MHz and is designed for very short-range communication, typically within 10 cm. It supports data rates from 106 kbps to 424 kbps and is widely used in contactless payments, ticketing, and access control due to its secure, proximity-based nature.

Infrared (IR)

Infrared communication uses infrared light for short-range transmission, typically up to 5 meters. It's used in remote controls and simple data transfers. Data rates typically max out at 4 Mbps, and it requires a direct line of sight, making it ideal for basic control applications.

Satellite

Satellite communication provides global coverage through orbiting satellites, utilizing frequency bands such as L-band, C-band, and Ka-band. While offering global reach, it has higher latency (especially with geostationary satellites). It's used for telecommunications, GPS, and broadcasting.

Cellular Networks (GSM, CDMA, 3G, 4G)

Cellular technologies have evolved from GSM (14.4 kbps) and CDMA (up to 2 Mbps) to 3G (up to 14 Mbps) and 4G (up to 1 Gbps). 4G provides high-speed mobile broadband, supporting applications like video streaming and online gaming. Cellular networks rely on frequency bands like 700 MHz to 2.6 GHz for communication.

Comparison Table

Parameter	Bluetooth	Wi-Fi	Zigbee (XB)	LoRa	5G	NFC	Infrared	Satellite	Cellular (GSM, CDMA, 3G, 4G)
Range of Communication	100 meters	100-150 meters	100 meters	Up to 15 km	1-2 km (urban)	10 cm	5 meters	Global coverage	1-50 km (depends on tech)
Cost	Low (₹100 - ₹500)	Moderate (₹500 - ₹2,500)	Low (₹200 - ₹1,500)	Low (₹500 - ₹4,000)	High (₹15,000 - ₹50,000)	Low (₹50 - ₹500)	Low (₹50 - ₹500)	High (₹20,000 - ₹1,00,000)	Moderate to high (500 - ₹2,000)
Propagation Delay	Low	Moderate	Low	High	Very low (1-10 ms)	Very low	Moderate	High (500 ms)	30-100 ms (depends on tech)
Power Consumption	Low (BLE)	Moderate	Very low	Very low	Moderate to high	Very low	Low	High	Varies (energy-efficient in 4G)
Throughput	1-3 Mbps	54 Mbps to 9.6 Gbps	20-250 kbps	0.3-27 kbps	100 Mbps to 10 Gbps	Up to 424 kbps	Up to 4 Mbps	1-100 Mbps	56 kbps (GSM) to 1 Gbps (4G)